

Test Plan Document — Bank Management System

Covers FR-001 through FR-005. Each module includes Unit (UT), Integration (IT) and System (ST) level test cases.

Login & Authentication (FR-001)

Test Case ID	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_01	Valid login with correct credentials	User is a registered customer	1. Open the app 2. Enter valid username 3. Enter valid password 4. Tap Login	Username: rahul_k Password: Bank@123	Dashboard loads with correct balance within 2 seconds	Dashboard loaded in 1.6s, balance correct	Pass
UT_02	Login attempt with wrong password	Account exists	1. Enter valid username 2. Enter wrong password 3. Tap Login	Username: rahul_k Password: wrongpass1	Error shown: "Incorrect username or password", no session created	Matched expected result	Pass
UT_03	Login with an unregistered username	None	1. Enter a username that doesn't exist 2. Enter any password 3. Tap Login	Username: ghost_user Password: any123	Same generic error as wrong password, no account info revealed	Matched expected result	Pass
UT_04	Password field masks input	None	1. Tap password field 2. Type password	Password: pass123	Characters appear masked as dots	Matched expected result	Pass
IT_01	Dashboard pulls balance after successful login	Valid credentials available	1. Log in successfully 2. Observe dashboard load	Username: rahul_k Password: Bank@123	Balance shown matches the value in the account service	Balance was off by ₹1 due to a caching delay	Fail
IT_02	Session expires after inactivity	User logged in	1. Log in 2. Leave the app idle for 15 minutes 3. Try to perform an action	No input for 15 minutes	Session expired message shown, user redirected to login	Matched expected result	Pass
ST_01	Login performance under concurrent load	50 test accounts available	1. Trigger 50 simultaneous login requests 2. Record response times	50 valid username/password pairs	All 50 logins complete within 2 seconds each	47 of 50 completed within 2s, 3 took up to 2.4s	Fail
ST_02	Account locks after repeated failed logins	Account exists, unlocked	1. Enter wrong password 5 times in a row	Password: wrongpass1 (x5)	Account locked for 10 minutes with a warning message	Matched expected result	Pass

Deposit & Withdrawal (FR-002)

Test Case ID	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_05	Deposit a valid amount	Active account, logged in	1. Select Deposit 2. Enter amount 3. Confirm	Amount: ₹5,000	Balance increases by ₹5,000, confirmation shown	Matched expected result	Pass
UT_06	Attempt to deposit a negative amount	Active account, logged in	1. Select Deposit 2. Enter a negative amount 3. Confirm	Amount: -₹500	Error shown: "Enter a valid amount"	Matched expected result	Pass
UT_07	Withdraw an amount within balance	Balance $\geq$ withdrawal amount	1. Select Withdraw 2. Enter amount 3. Confirm	Balance: ₹10,000 Withdraw: ₹2,000	Balance reduces to ₹8,000, confirmation shown	Matched expected result	Pass
UT_08	Withdraw more than the available balance	Balance < withdrawal amount	1. Select Withdraw 2. Enter amount greater than balance 3. Confirm	Balance: ₹8,000 Withdraw: ₹20,000	Error: "Insufficient funds", balance unchanged	Matched expected result	Pass
IT_03	Notification sent after a deposit	Deposit completed	1. Complete a deposit 2. Wait for SMS/email alert	Amount: ₹3,000	Notification received within 1 minute	Notification arrived after 4 minutes due to a queue backlog	Fail
IT_04	Ledger updates after withdrawal	Withdrawal completed	1. Complete a withdrawal 2. Check transaction ledger	Amount: ₹1,500	Ledger entry created instantly	Matched expected result	Pass
ST_03	Simultaneous deposit and withdrawal on same account	Same account used by two requests	1. Trigger a deposit and a withdrawal at the same moment 2. Check final balance	Deposit ₹2,000 + Withdraw ₹1,000	Final balance reflects both correctly, no race condition	Balance was correct, 0.8s extra delay observed	Pass
ST_04	Response time under concurrent transaction load	100 accounts available	1. Trigger 100 concurrent deposit/withdrawal requests 2. Record response times	100 mixed transactions	95% of requests complete within 3 seconds	96% completed within 3 seconds	Pass

Fund Transfer (FR-003)

Test Case ID	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_09	Transfer to a valid existing account	Sender has sufficient balance	1. Enter receiver account number 2. Enter amount 3. Confirm transfer	Receiver: 12345 Amount: ₹1,000	Sender debited, receiver credited correctly	Matched expected result	Pass
UT_10	Transfer to a non-existent account	None	1. Enter an invalid account number 2. Enter amount 3. Confirm transfer	Receiver: 00000 Amount: ₹500	Error: "Recipient account not found"	Matched expected result	Pass

Test Case ID	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_11	Transfer amount exceeding sender's balance	Balance < transfer amount	1. Enter receiver account 2. Enter amount greater than balance 3. Confirm	Balance: ₹10,000 Transfer: ₹50,000	Error: "Insufficient balance", no debit occurs	Matched expected result	Pass
UT_12	Transfer to the sender's own account	None	1. Enter the sender's own account number as receiver 2. Enter amount 3. Confirm	Same account number on both sides	Error: "Cannot transfer to the same account"	System allowed the transfer with no warning shown	Fail
IT_05	Both balances update within a single transaction	Two valid accounts	1. Complete a transfer 2. Check both account balances	Transfer: ₹2,000	Sender and receiver balances update atomically	Matched expected result	Pass
IT_06	Rollback on failure mid-transfer	Simulated network drop mid-transfer	1. Start a transfer 2. Force a network failure before completion 3. Check sender balance	Transfer: ₹5,000	Debited amount is automatically rolled back to sender	Rollback happened, but took about 12 seconds	Pass
ST_05	Concurrent transfers on the same pair of accounts	Two accounts, multiple simultaneous transfers	1. Trigger several transfers between the same two accounts at once 2. Check final balances	Multiple ₹500-₹1,000 transfers	No data corruption, correct final balances on both sides	Slight lag but final balances were correct	Pass
ST_06	Transfer entries appear in transaction history	Transfer completed	1. Complete a transfer 2. Check transaction history for both accounts	Transfer: ₹1,000	Entry visible in history for both sender and receiver immediately	Matched expected result	Pass

Loan Management (FR-004)

Test Case ID	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_13	Submit a loan application with valid details	Customer logged in	1. Fill loan form 2. Enter amount and tenure 3. Submit	Amount: ₹2,00,000 Tenure: 24 months	Application submitted, reference ID generated	Matched expected result	Pass
UT_14	Submit application with a required field missing	Customer logged in	1. Fill loan form leaving income field blank 2. Submit	Income field left empty	Form blocks submission, highlights the missing field	Matched expected result	Pass
UT_15	Submit a loan amount exceeding the eligible limit	Customer logged in	1. Enter an amount above the allowed maximum 2. Submit	Amount: ₹50,00,000	Error: "Amount exceeds eligible limit"	Matched expected result	Pass

Test Case ID	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
IT_07	Submitted application reaches officer's queue	Application submitted	1. Submit application 2. Check loan officer's queue	Reference ID from submission	Application appears in queue within a few seconds	Appeared after roughly 8 seconds	Pass
IT_08	Officer approves an application	Application in queue	1. Officer opens application 2. Approves it	Application reference ID	Customer notified of approval, status updated to Approved	Matched expected result	Pass
IT_09	Officer rejects an application with a reason	Application in queue	1. Officer opens application 2. Rejects with a reason 3. Submits decision	Reason: "Insufficient income proof"	Customer notified with the rejection reason included	Notification sent, but the reason field was blank	Fail
ST_07	Multiple customers submit applications at the same time	Several customers logged in	1. Trigger simultaneous loan submissions 2. Check reference IDs generated	5 concurrent applications	Each application receives a unique reference ID	Matched expected result	Pass
ST_08	Full loan lifecycle from application to notification	None	1. Customer applies 2. Officer reviews 3. Officer approves 4. System notifies customer	End-to-end single application	Complete flow works without manual steps beyond review	Matched expected result	Pass

Transaction History (FR-005)

Test Case ID	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_16	View transactions from the last 12 months	Account has transaction history	1. Open Transaction History 2. View list	Account with 12 months of activity	All transactions listed with date, amount and type	Matched expected result	Pass
UT_17	Filter history by transaction type	Account has mixed transaction types	1. Open Transaction History 2. Apply filter for one type	Filter: Withdrawals only	Only matching transaction type is shown	Matched expected result	Pass
UT_18	View history for an account with no transactions	New account, no activity	1. Open Transaction History	New account, zero transactions	Empty-state message shown instead of a blank screen	Matched expected result	Pass
IT_10	New transaction appears in history right after it happens	A transaction has just been completed	1. Complete a transaction 2. Immediately open Transaction History	One recent deposit	New entry appears in history instantly	2-3 second delay observed under load	Pass
UT_19	Auditor cannot view another customer's history without authorization	Auditor logged in, no explicit access granted	1. Log in as Auditor 2. Attempt to open a customer's history without authorization	Auditor role, unauthorized account	Access denied unless proper authorization is granted	Matched expected result	Pass
IT_11	Long transaction lists paginate correctly	Account with 100+ transactions	1. Open Transaction History 2. Scroll through pages	Account with 120 transactions	20 transactions load per page correctly	Matched expected result	Pass

Test Case ID	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
ST_09	Transactions are sorted with the latest first	Account with multiple transactions on different dates	1. Open Transaction History 2. Check the order of entries	Transactions across 3 different dates	List is sorted most recent to oldest	Matched expected result	Pass
ST_10	Export transaction history as a statement	Account has transaction history	1. Open Transaction History 2. Tap Export/Print	Account with 6 months of history	Correctly formatted statement is generated	Export worked, but date format was inconsistent	Fail